



Численное моделирование биотеплопередачи при криоабляции опухолей правильной формы в тканях печени с использованием мультizonдов

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Основные моменты

- Изучается возможность криохирургического лечения множественных опухолей в тканях печени.
- С помощью численного моделирования можно спрогнозировать количество криозондов, их размер и время работы.
- При замораживании опухолевых тканей происходит фазовый переход из жидкого состояния в твердое.
- Важными факторами являются захват фронта перехода и размер образующихся ледяных шариков.

- Моделирование проводилось с помощью метода конечных элементов в COMSOL 5.5, результаты были проверены.

Абстрактный

Настоящее исследование посвящено стратегии предоперационного планирования криохирургического лечения множественных опухолей правильной формы в трехмерной структуре ткани печени. Численное моделирование позволяет определить оптимальное количество криозондов, их расположение, время операции и степень термического некроза опухоли и окружающих здоровых тканей. Для эффективной криохирургии необходимо поддерживать в опухолевых клетках температуру от -40°C до -50°C , смертельную для них. Процесс замораживания нежелательных опухолевых тканей включает в себя фазовый переход от жидкой фазы методом конечных элементов теплоемкости к твердой фазе, точное определение фронта перехода и размера или расположения ледяных шариков, образующихся в процессе, являются важными факторами криохирургии. В этом исследовании был использован метод фиксированной области для учета скрытой теплоты фазового перехода в уравнении биотеплоп. Были проанализированы ледяные шарики, полученные с помощью разного количества зондов. Численное моделирование проводилось с использованием стандартного пакета COMSOL 5.5, а полученные результаты были сопоставлены с результатами предыдущих исследований.

Введение

По данным Всемирной организации здравоохранения (ВОЗ), в 2020 году рак стал основной причиной смерти, унеся жизни почти 10 миллионов человек. В литературе описаны такие методы лечения рака, как химиотерапия, лучевая терапия, хирургическое иссечение, радиочастотная абляция, криохирurgia и так далее. Криохирurgia — один из эффективных методов лечения рака, который предпочтительнее традиционных методов, поскольку он малоинвазивный, менее затратный и имеет хорошие прогнозы (Кумар и Рай, 2021; Чуа, 2011; Чуа, 2013).

Криохирurgia — это хирургическая операция, при которой пораженные раком ткани разрушаются путем нанесения на них небольшого разреза и воздействия экстремально низкой температуры с помощью криозондов. В качестве криозондов обычно используются твердый углекислый газ, оксид азота, аргон

и жидкий азот. Одним из наиболее эффективных криогенов является жидкий азот, который кипит при температуре -196°C при нормальном атмосферном давлении. Продолжительность и степень замораживания являются предметом исследований. Купер (1964) первоначально заявил, что поддержание ткани при температуре ниже -20°C в течение одной минуты вызывает разрушение клеток. Позже смертельная температура была пересмотрена до -60°C Нилом (Neel et al., 1971) и -70°C Стареном и др. (1997). Разрушение клеток начинается при температуре от -20°C до -30°C , но оно не является определенным, пока не достигнет -40°C . Тем не менее на основании некоторых данных, полученных в ходе исследований *in vitro*, а также клинических экспериментов, принято считать, что продолжительность воздействия не имеет значения, если ткань хранится при температуре от -40°C до -50°C или ниже, что обеспечивает полное разрушение клеток (Chua, 2011; Liu et al., 2003; Baust et al., 2014; Patil, 2021).

Inappropriate freezing of tumour cells may not cover the entire harmful tumour cells or may destroy the adjacent healthy tissue which will both results in failure of the treatment. Researchers have worked on ways to enhance the cryosurgery procedure to achieve complete destruction of abnormal tumour cells in the mean time imposing no significant injury to the surrounding healthy tissue. Rewcastle et al. (2001) used multiprobes to develop time dependent three dimensional model using finite difference method. Liu et al. (2003) studied iceball formation using multiprobes based on Finite Element Method. Rabin and Stahovich (2003) have employed cryo heater to shape frozen region which enhanced the process of cryosurgery. Chua (2011) performed an in-vitro experiment considering a single irregular large tumour and studied the effect of different probe sizes and positions. Irregular shape of tumours might be a challenge to the treatment as the iceball produced may not completely put the entire regions of the irregular shapes under lethal temperature or may damage the healthy tissue in the process of freezing. In case of multiprobes, position of the probes will be a great factor for successful operation. In view of that, Jaberzadeh and Essert (2015) did a comparison of different optimization method for pre operative planning of 3D liver cryosurgery using multiprobes. Nabaei and Karimi (2018) had investigated the effect of vessel size and distance from a single liver tumour and found that large artery can significantly affect the cryosurgery outcome. They had also injected nanoparticles to enhance the process. Shurrab et al. (2016) discussed a two phase model investigating temperature distributions inside and outside the multiprobes and the cooling effect of the cryoprobes which is a valuable factor for the efficiency of the surgery. Nazemian and Nabaei (2020) studied the effect of vascular network complexity on liver tumour placing a single probe in different positions of the liver tissue. Cryoablation coverage rate and iceball irregularity were analysed. Gunjal et al. (2022b) developed a four staged numerical

methodology for efficient cryoprobe placement inside an irregular tumour using superimposed standard volume. They have also tracked the freezing front and the associated temperature distribution involved in cryosurgery using lensless Fourier transform digital holographic interferometry with water and agar biogel as their working medium (Gunjal et al., 2022a).

Eventhough the cryosurgery procedure can be termed simple, it experiences a phase transition when temperature falls in the range of -1°C to -8°C (Nazemian and Nabaei, 2020; Deng and Liu, 2006; Tanwar and Kumari, 2020). The fixed domain method (effective heat capacity method) and front tracking method are one of the commonly used methods for simulating this phase transition problem. In case of front tracking method, the whole domain is divided into liquid and solid domain where the moving boundary is considered to be the latent heat. While in fixed domain, the domain is considered to be a single domain and position of interfaces are known. Fixed grid method have the advantage over front tracking method as it can be implemented using computer codes easily and is suitable for materials having finite freezing range. Various numerical techniques have been proposed to tackle this phase change problem as discontinuity might happen in the transition front. Finite Volume method (Kudryashov and Shilnikov, 2015; Das and Mishra, 2015), finite difference method (Kumar and Rai, 2021; Patil and Maniyeri, 2019), finite element method (Nazemian and Nabaei, 2020; Yang et al., 2008), Extended finite element method (X-FEM) (Tanwar and Kumari, 2020), hybrid method (Singh and Bhargava, 2015; Condino et al., 2021), etc are the various techniques available in literature to solve such phase transition problem.

A successful cryosurgical treatment requires an optimal planning in terms of number of cryo-probes to be used, duration of freezing and placement of cryo-probes, etc. to destroy all the cancerous cells while giving minimum damage to healthy adjacent tissues. Numerous studies have supported that cryosurgery is an effective treatment for liver cancer (Niu et al., 2014). Single regular tumour in liver tissue have been considered widely (Nabaei and Karimi, 2018; Nazemian and Nabaei, 2020). In practical, the cancer cells may have metastasized causing irregular shape and multiple tumours. To the best of the author's knowledge, there is limited study available for cyrosurgical simulation of multiple tumours within a single tissue. This study focuses upon numerical simulation for cryo-ablation of multiple tumours in a liver tissue. Size, number of cryoprobe and its active length required in the cryosurgical process has been investigated in detail. Finite Element Method software called COMSOL is used to obtain the temperature distribution across the liver domain. Necrosis of the healthy surrounding tissue is also analysed. The numerical results may offer significant guidance for pre-planning of the procedure and could be of medical assistance to the professionals with real medical tumour images.

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Model geometry

In this study, 3D Liver image from grabCAD is considered for simulation which has an irregular computational domain illustrated in Fig. 1, size of the tissue is approximately 15cm, 13cm, 10cm in x, y, z direction respectively. This study consists of cryosurgical simulation of three spherical tumours of 2cm diameter each located at different positions given in Table 1. Cylindrical cryoprobes of 3mm, 6mm and 8mm diameters have been considered. The active length has been taken as 10mm, ...

Results and discussion

In this study, cryosurgery process is simulated for a liver cancer tissue with three spherical tumours. To numerically investigate the ideal active part involved in the cryosurgery process, a single cryoprobe in a single tumour has been considered. The probe diameter is varied from 3mm to 6mm. It can be seen from Table 3 that, cryoprobe of 6mm diameter with 16mm active part is sufficient to bring the entire tumour boundary (i.e 10mm from centre of the probe) under lethal temperature of -40°C ...

Conclusion

In this study, a 3D liver undergo cryosurgery of regular multiple tumours which are assumed to be equal in dimension. A single cryoprobe of 3mm diameter was first considered for the simulation, however, it is largely insufficient to keep the whole tumours under lethal temperature. The probe diameter was varied to 8mm which does not give the desired results either. Finally, multiprobe of 6mm diameter is considered placing the three probes at the centre of each tumours and the temperature ...

CRediT authorship contribution statement

Sonam Tanwar: Conceptualization, Methodology, Supervision. **Lalhmingsangi Famhawite:** Software, Writing – original draft. **Pooja Raj Verma:** Writing – review &

editing. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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...However, their investigations do not enunciate the effects of blood vessel during cryo-treatment with two-phase thermo-fluid approach [22,23]. Besides, the utilization of multiple cryoprobes is

employed in the context of cryo-surgical treatment case to enhance the outputs of the synergistic effects where tumors or lesions exhibit intricate shapes or irregular boundaries [24,25]. This technique ensures comprehensive coverage of the entire target area with sufficient freezing, thereby reducing the risk of recurrence when the placement of cryoprobes into the tumor is crucial for cryosurgery [26]...

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